

Mrigesh Thakur

Github: Legend101Zz
Linkedin: mrigesh-thakur-11new
Portfolio: mrigeshthakur-portfolio.netlify.app

Email: mrigeshthakur11@gmail.com
Mobile: +91-85807-32070

EDUCATION

- National Institute of Technology** Himachal Pradesh, India
Bachelor of Technology in Mathematics and Scientific Computing; GPA: 9.18
Dec 2021 - Jul 2025
- Cambridge International School** Himachal Pradesh, India
Higher Secondary; Percentage: 95.4
Apr 2011 - May 2021

SKILLS

- Languages:** JavaScript, TypeScript, Python, Rust, C++
- Frameworks:** React, Express, Backtesting.py, Three.js, Cython
- Skills:** Algo Trading, Data Structures, Computer Networking, JIT Compilation
- Tools:** Mongo ATLAS, WebRTC, WebSockets, Git, Adobe Creative Suite, Final Cut Pro, Blender
- Soft Skills:** Leadership, Event Management, Writing, Public Speaking, Time Management

EXPERIENCE

- Google Summer of Code - INCF** Remote
Open Source Developer May 2025 - August 2025
 - Brian Simulator JIT Replacement:** Replacing Brian’s JIT compilation mechanism, migrating from Cython to modern alternatives. Implementing C++ structures for improved performance and exploring Numba/SciPy Weave for enhanced runtime efficiency.
- Walmart** Bangalore, India
Data Science Intern May 2024 - August 2024
 - Delivery Date Prediction:** Predicted delivery dates for Walmart Mexico using **LightGBM** and custom asymmetric loss function, impacting 5M+ users with improved accuracy.
- Summer of Bitcoin Fellowship - Caravan@Unchained Capital** Remote
Open Source Developer April 2024 - September 2024
 - Fees Package:** Implemented modular BTC fee package for Caravan, enhancing multisig fee bumping strategies (RBF, CFPF).
- Meta’s XR Open Source Fellowship - Fanisko** Remote
Open Source Developer May 2023 - Jul 2023
 - AR Applications:** Developed **3 AR applications** during fellowship. AR cricket wagon wheel used in **ICC’s ODI World Cup** app with **1M+ users**.

PROJECTS

- Algame-MRIMADHA: (Published) PyPI**
Modern algorithmic trading & backtesting framework with GUI interface. Provides flexible strategy development, robust testing, and professional-grade analysis tools.
Tech: Python, Backtesting.py, Pandas, NumPy, Matplotlib
- Caravan Regtest Manager: (Published) NPM**
Terminal utility for managing Caravan multisig wallets in regtest mode. Simplifies Bitcoin development with wallet management, PSBT handling, and block mining.
Tech: Node.js, Typescript, Bitcoin Core, Multisig, PSBT
- Unified Liquidity Protocol NFTs: (Completed) GitHub**
Tokenized cross-chain liquidity positions as NFTs, enabling seamless liquidity management. Built smart contracts and marketplace with Euclid protocol integration.
Tech: Rust, CosmWasm, IBC Protocol, React.js, TypeScript, Archway blockchain
- Appy: (In Progress) Pitch Deck**
Automated apple harvesting using IoT, ML-based grading, and gentle shaking mechanism. Achieved efficient plucking and sorting while preserving quality.
Tech: IoT sensors, ML/AI, Image processing
- NITH-Portal: (Approved) GitHub**
Digitalized outpass system for **4000+ students**, saving **500+ admin hours/year**. Integrated biometric authentication in mobile app.
Tech: Flutter, NodeJS, ReactJS, MongoDB

ACHIEVEMENTS

- Scholar:** Google Summer of Code 2025 - INCF.
- Scholar:** Summer of Bitcoin 2024 - Caravan@Unchained
- Scholar:** Meta’s XR Open Source Fellowship 2023 - Fanisko
- Incubated:** Project ”Appy” - Incubated under HP CM-Startup Scheme at NIT-Hamirpur with **5,000 USD** funding.
- Published:** Research paper ”Adaptive Strategy Selection Framework Using Hidden Markov Models for Market Regime Detection” in Computational Economics Journal.
- Best Research Paper:** ABHIGYA 2023 for project Appy
- Winner:** Build the Future of DeFAI - Sonic Blockchain: Won **7,500 USD** for project KnowScroll.
- Winner:** Archway Hunt-A-Thon 2024: Won **5,000 USD** from **100,000 USD** prize pool for project Deauth
- Winner:** Euclid Aurora Hackathon 2024: Won **10,000 USD** from **100,000 USD** prize pool
- Winner:** Andromeda’s Stellarscape Explorer Hackathon 2024: Secured **2,000 USD** from **10,000 USD** prize pool.
- Winner:** 1st place from **10,000 teams participating** - Walmart’s Sparkathon 2023.
- Runner Up:** 2nd place - Findoc’s Algo-Ninja 2023: Won **2,000 USD** from **12,000 USD** prize pool.